Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] **[18DS5DE11]**

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

DATAWAREHOUSING AND DATA MINING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. With a neat diagram, explain the multilayer architecture of Datawarehousing. **07**

b. What is fact table and explain the types of fact tables in data warehousing **07**

2. a. Suppose that the data for analysis includes attribute age. The age values for data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 23, 29, 35, 41, 45, 53, 62, 69, 72. Apply z-score transformation to transform the value 45 for age onto the range [0,1]. **07**

b. What is data pre-processing? What are the techniques available in data pre-processing? **07**

3. a. List and explain factors affecting the computational complexity of Apriori algorithm. **04**

b. Consider the following training dataset, **10**

TID Items
T1 {M,O,N,K,E,Y}
T2 {D,O,N,K,E,Y}
T3 {M,A,K,E}
T4 {M,U,C,K,Y}
T5 {C,O,O,K,I,E}

Construct FP tree. Show the trees separately after reading each transaction and also find frequent item-set.(Given minsupp=3)

OR

4. a. Explain about FP growth algorithm and Mention the steps involved in FP growth Algorithm. **07**

b. Discuss about Association rule mining algorithm with an example **07**

5. a. Create confusion matrix and find: **07**

a) Accuracy b) Precision c) Recall d) F1-Score

for the following dataset

Actual values = ['dog', 'cat', 'dog', 'cat', 'dog', 'dog', 'cat', 'dog', 'cat', 'dog']

Predicted values = ['dog', 'dog', 'dog', 'cat', 'dog', 'dog', 'cat', 'cat', 'cat', 'cat']

b. Explain KNN algorithm with suitable example. **07**

OR

6. a. Consider the following set of training example **10**

X	Y	Z	No. of class C1 example	No. of class C2 example
0	0	0	5	40
0	0	1	0	15
0	1	0	10	5
0	1	1	45	0
1	0	0	10	5
1	0	1	25	0
1	1	0	5	20
1	1	1	0	15

Compute a two-level decision tree. Use the classification error rate as the criterion for splitting. What is the overall error rate of the induced tree?

b. List the characteristics of Nearest Neighbour classifier. **04**

7. a. What is hierarchical clustering and explain the types of hierarchical clustering? **07**

b. Discuss in detail about K-means cluster algorithm with an example. **07**

OR

8. a. Perform Agglomerative Algorithm on the following data and plot a dendogram using single link approach. The given data indicates distance between elements. **10**

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

b. What is clustering? List and explain different types of clusters. **04**